

A PROJECT TITLE

ON

Heart Disease Prediction Using Ridge- and Lasso-Regularized Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

Fathima Qhibtiya Khader 2520030155

Pakalapati Harini 2520030177

Under the Esteemed guidance of

Shaik Asif

Assistant Professor

CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



The project utilizes a publicly available heart disease dataset (commonly sourced from the UCI Machine Learning Repository / Kaggle, based on the Cleveland heart disease data), which contains patient attributes such as age, sex, chest pain type, blood pressure, cholesterol, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, and other diagnostic measurements, along with a binary label indicating the presence or absence of heart disease. Data preprocessing steps, including handling missing values, encoding categorical variables, and feature scaling, are applied prior to model training so that the regularization penalties act fairly across all features.

The models are implemented using Python and evaluated using performance metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC, with the regularization strength tuned via cross-validation. The proposed system aims to provide an accurate, interpretable, and computationally efficient tool for early screening of cardiovascular risk, while comparing the trade-offs between Ridge and Lasso regularization in a clinical prediction setting.

3. SOFTWARE REQUIREMENTS

The software requirements are listed below.

Operating System

- Windows 10/11, Mac or Linux

Programming Language

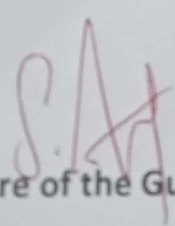
- Python
- Libraries- NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

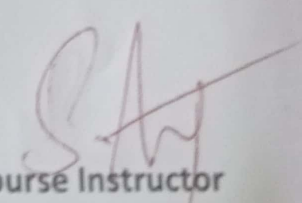
Development Environment

- Visual Studio Code (VS Code)
- Google Colab

Dataset

- Heart Disease Dataset (UCI Machine Learning Repository / Kaggle, .csv format)
<https://www.kaggle.com/datasets/redwankarimsony/heart-disease-data>


Signature of the Guide


Course Instructor